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1. Which of the following are the assumptions of the optical flow constraint equation?

1 / 1 point

- ☐ Brightness Constancy
- ☐ Small Motion between small intervals of time
- ☐ Nearby points in the image plane move in a similar manner
- ☒ All of the above

 **Correct**

a) Brightness Constancy: This assumption states that the brightness of a point in an image remains constant as it moves over time. This is a fundamental assumption in optical flow estimation.

b) Small Motion between small intervals of time: This assumption suggests that the motion of objects in the scene is relatively small between consecutive frames of an image sequence.

c) Nearby points in the image plane move in a similar manner: This assumption implies that points that are close to each other in the image plane will have similar optical flow vectors. In other words, neighboring pixels in the image will move together in a coherent manner.

These assumptions are crucial for the optical flow constraint equation to work effectively, as violations of these assumptions can lead to inaccuracies in optical flow estimation.

2. For a given scene, the spatial gradients of image intensity are $I_x = 2$, $I_y = 3$, and $I_t = 4$. Which one of the following values of u and v satisfies the optical flow constraint equation?

1 / 1 point

- ☐ $u = -1; v = 2$
- ☒ $u = 1; v = -2$
- ☐ $u = -2; v = 1$
- ☐ $u = 2; v = -1$

 **Correct**

The optical flow constraint equation is $u \cdot I_x + v \cdot I_y + I_t = 0$

Putting the values given in the question.

$$2u + 3v + 4 = 0$$

This implies, $u = - (4+3v)/2$

We need to find a (u,v) pair such that the above equation is satisfied.

Only the option B) satisfies it as follows.

$$u = - (4 + 3 \cdot (-2))/2 = - (4 - 6)/2 = 2/2 = 1$$

3. What is the aperture problem in the context of computer vision and motion estimation?

1 / 1 point

- ☐ A problem with camera lenses affecting image quality
- ☒ The challenge of estimating motion when only a limited view of an object's motion is available
- ☐ A problem related to image compression techniques
- ☐ A limitation of motion capture technology

 **Correct**

The aperture problem describes a difficulty in accurately perceiving the movement of an object through a limited observation window (aperture).